

2020 AMC 10B Problem 4 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 4

Problem:

The acute angles of a right triangle are a° and b° , where $a > b$ and both a and b are prime numbers. What is the least possible value of b ?

Answer Choices:

- A. 2
- B. 3
- C. 5
- D. 7
- E. 11

Solution:

Since the three angles of a triangle add up to 180° and one of the angles is 90° because it's a right triangle,

$$a^\circ + b^\circ = 90^\circ$$

The greatest prime number less than 90 is 89. If $a = 89^\circ$, then

$$b = 90^\circ - 89^\circ = 1^\circ$$

which is not prime.

The next greatest prime number less than 90 is 83. If $a = 83^\circ$, then $b = 7^\circ$, which is prime, so we have our answer:

$$\boxed{(D)7}$$

OR

Looking at the answer choices, only 7 and 11 are coprime to 90. Testing 7, the smaller angle makes the other angle 83 which is prime, therefore our answer is $\boxed{(D)7}$.

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